In-Class Activity 2 — Use Case and Activity Diagrams

CIS 641 Systems Analysis and Design · Fall 2026 · Dr. Sutrave

Term project: PersonaSync™ — Phase 3, context delivery

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Figure 1. Use Case Diagram 1 — Persona and context management (as-is web app)

Figure 2. Use Case Diagram 2 — Context delivery at the point of use (to-be Phase 3 client)

Figure 3. Activity Diagram 1 — Deliver context into an AI session

Figure 4. Activity Diagram 2 — Update context from inside a session

Use Case Diagram 1 — Persona and context management

How a user builds and maintains their declared context in the PersonaSync web app

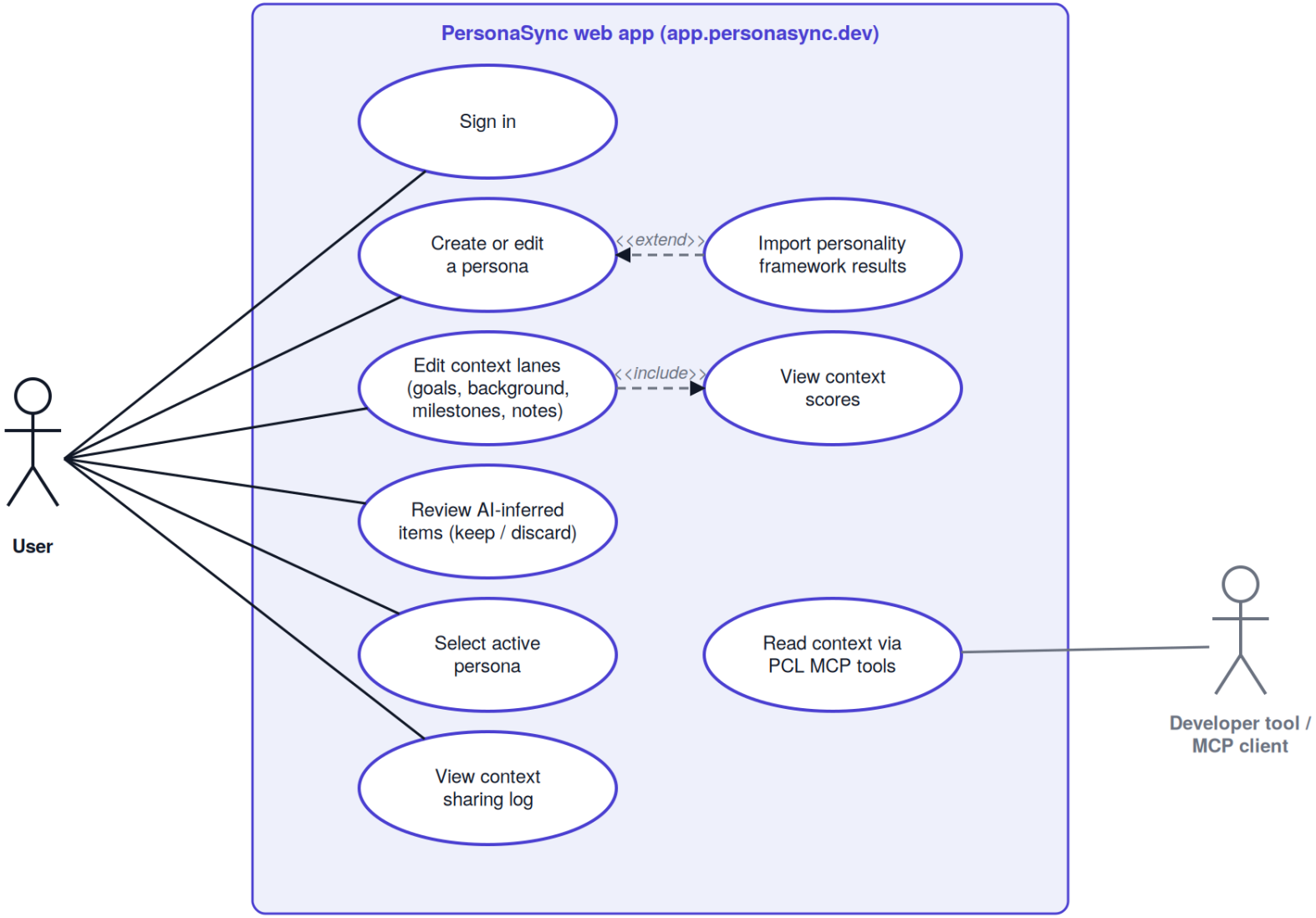


Figure 1. Use Case Diagram 1 — Persona and context management. The as-is system at app.personasync.dev. The User signs in, creates or edits a persona, edits the four context lanes (goals, background, milestones, notes), reviews AI-inferred items to keep or discard them, selects the active persona, and views the sharing log. Importing personality framework results is an optional extension of creating a persona; viewing context scores is included whenever lanes are edited. A developer tool or MCP client is a secondary actor that reads context through the Personal Context Layer (PCL) MCP tools.

Use Case Diagram 2 — Context delivery at the point of use

Phase 3: carrying stored context into the AI tools where the user actually works

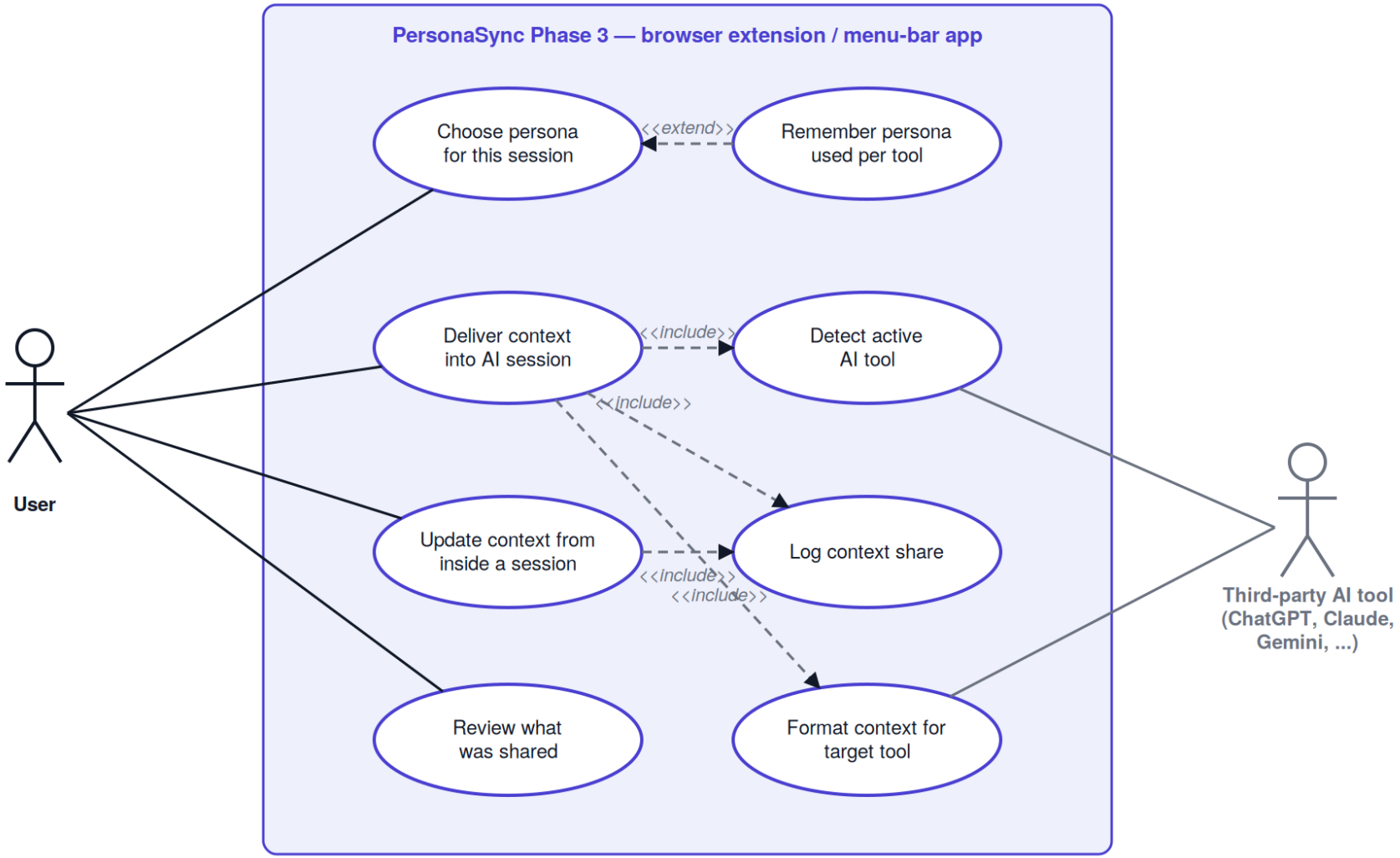


Figure 2. Use Case Diagram 2 — Context delivery at the point of use. The to-be Phase 3 client (browser extension or macOS menu-bar app). The User chooses a persona for the session, delivers context into the AI session, updates context from inside a session, and reviews what was shared. Delivering context always includes detecting the active AI tool, formatting the context for that tool, and logging the share; in-session updates are also logged. Remembering the persona used per tool is an optional extension of choosing a persona. The third-party AI tool (ChatGPT, Claude, Gemini, etc.) is a secondary actor involved in detection and formatting.

Activity Diagram 1 — Deliver context into an AI session

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Main flow for the Phase 3 extension: from opening a tool to a logged, context-loaded session

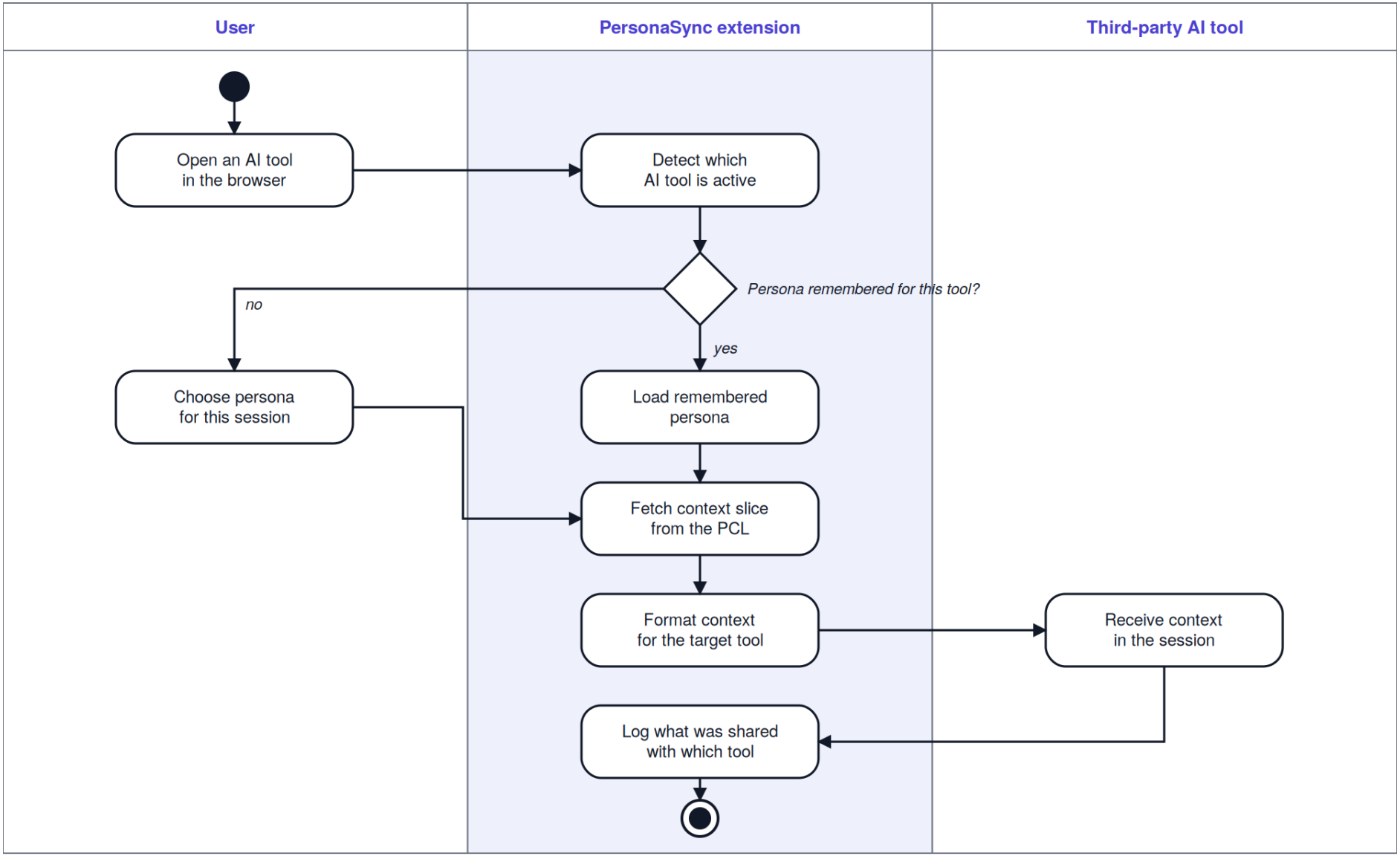


Figure 3. Activity Diagram 1 — Deliver context into an AI session. The main flow behind the “Deliver context” use case. The User opens an AI tool; the extension detects which tool is active and checks whether a persona is remembered for it. If not, the User chooses one. The extension then fetches the context slice from the PCL, formats it for the target tool, the AI tool receives it in the session, and the extension logs what was shared with which tool.

Activity Diagram 2 — Update context from inside a session

How an AI-inferred item is captured, reviewed, and either kept as declared context or discarded

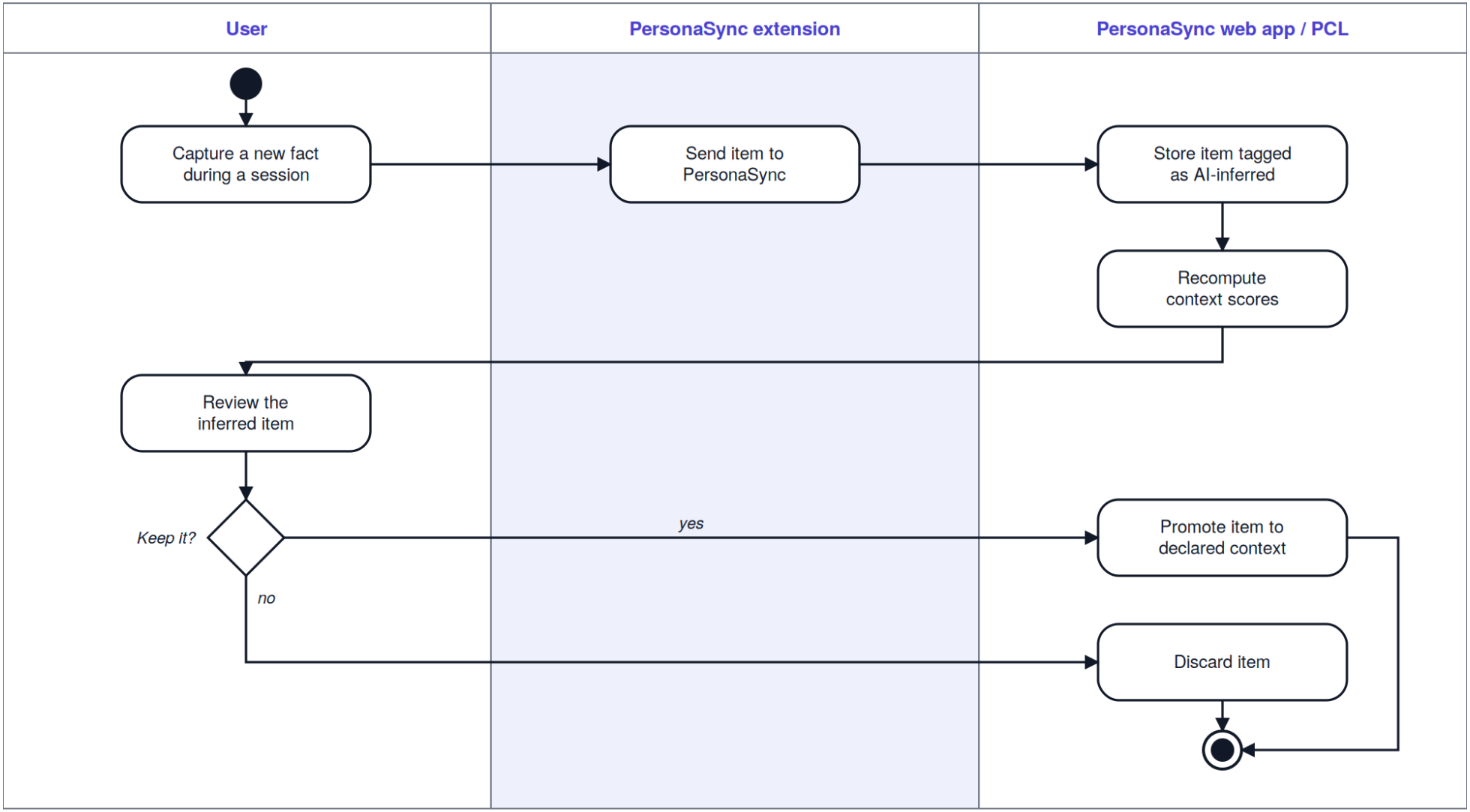


Figure 4. Activity Diagram 2 — Update context from inside a session. The flow behind the “Update context from inside a session” use case, and the “declared, not inferred” rule as a process. A fact captured mid-session is sent to PersonaSync and stored tagged as AI-inferred; scores recompute; the User reviews the item and decides whether to keep it. Kept items are promoted to declared context; rejected items are discarded.